

# When the Monsoon Satisfies: Extreme Weather, Agricultural Exposure, and Climate Awareness in India

APEP Autonomous Research\* @ailscl

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## Abstract

Does extreme weather make people care about climate change? I study this question in India, where 600 million people depend on agriculture and weather has first-order economic consequences. Combining state-monthly Google Trends data on climate-related searches with NASA weather records across 22 states (2004–2023), I find suggestive evidence that the effect of temperature anomalies on climate search interest depends on agricultural structure. In states with low crop dependence, warming anomalies are associated with increased climate search interest. In highly agricultural states, the same shocks are associated with *decreased* searches—the interaction of temperature with pre-period agricultural share is consistently negative across specifications. Wild cluster bootstrap inference, appropriate for the 22-state panel, indicates that this differential pattern is suggestive (WCB  $p \approx 0.10$ – $0.13$ ) rather than definitive at conventional significance levels. The findings are consistent with the hypothesis that economic immediacy may crowd out abstract climate concern in agricultural populations.

**JEL Codes:** Q54, D83, O13

**Keywords:** climate change beliefs, extreme weather, India, shift-share, Google Trends, agricultural exposure

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\*Autonomous Policy Evaluation Project. Correspondence: scl@econ.uzh.ch (cumulative: N/A).

# 1. Introduction

India lost \$87 billion to extreme weather events between 2000 and 2020, more than any country except China and the United States (Eckstein et al., 2021). The 2009 monsoon deficit—23 percent below normal—destroyed crops across two-thirds of the country. The 2023 heat wave pushed temperatures above 45°C in twelve states simultaneously. Yet India’s climate policy ambitions remain modest relative to its exposure, and individual awareness of climate change lags behind comparably affected nations (Pew Research Center, 2021). A natural question arises: if people are experiencing climate change firsthand, why doesn’t this translate more clearly into climate concern?

The existing literature on weather experience and climate beliefs offers a compelling answer for rich democracies: local temperature anomalies do shift beliefs, but only temporarily. Egan and Mullin (2012) show that a three-degree Fahrenheit above-normal weekly temperature increases the probability of acknowledging global warming by one percentage point among Americans—an effect that dissipates within twelve days. Deryugina (2013) finds similarly transient updating. This “recency bias” model implies that weather shapes attention, not conviction. But the evidence comes almost exclusively from the United States and Europe, where agriculture employs fewer than 3 percent of workers and weather is an inconvenience rather than an existential economic threat.

This paper asks whether the weather-to-beliefs channel operates differently where weather has first-order economic stakes. India provides an ideal laboratory. Over 40 percent of the labor force works in agriculture, and crop yields are tightly coupled to monsoon timing, temperature, and precipitation (Guiteras, 2009). Weather shocks are not abstract signals about planetary trends—they are income shocks that destroy livelihoods, drain savings, and trigger distress migration (Jayachandran, 2006). If experiencing a drought makes a farmer think “my rice crop failed” rather than “the planet is warming,” then the weather-to-beliefs channel could actually be *weaker* where weather matters most economically. This is the central hypothesis I test.

I construct a state-monthly panel of climate awareness for 22 Indian states from 2004 to 2023 using Google Trends search interest for “global warming” and “climate change.” I match this with daily temperature and precipitation data from the NASA POWER database, aggregated to state-month anomalies relative to 1981–2000 climate normals. To test the agricultural exposure mechanism, I interact weather anomalies with pre-period (triennium ending 2000) state-level shares of cultivated area under major crops, drawing on published agricultural statistics from the Indian Ministry of Agriculture and the ICRISAT District Level Database.

The empirical strategy exploits the plausible exogeneity of weather. Temperature and precipitation anomalies—deviations from long-run monthly means—are orthogonal to state-level economic and social characteristics conditional on state and time fixed effects. The primary specification regresses climate search interest on temperature anomalies with state and month-year fixed effects, clustering standard errors at the state level. The key test is the interaction of temperature anomalies with pre-period agricultural share: if economic exposure mediates the weather-beliefs channel, the interaction coefficient should be non-zero.

The main finding reveals substantial heterogeneity. In the baseline specification without interactions, temperature anomalies have a small, negative, and statistically insignificant effect on climate search interest ( $-0.29$ ,  $p = 0.18$ ). When I interact temperature with agricultural share, the coefficient on the bare temperature anomaly becomes positive ( $0.526$ ), while the interaction term is negative ( $-1.198$ ). In the log specification, both coefficients are larger in magnitude: temperature alone is associated with an 8.4 percent increase in climate searches, while the interaction implies a  $-17.1$  percent offset per unit of agricultural share. Evaluated at the median agricultural share of 0.70, the net effect of a one-degree warming is  $-3.6$  percent. However, wild cluster bootstrap inference appropriate for the 22-state panel yields  $p$ -values of 0.10–0.13 for the interaction, indicating that the differential pattern is suggestive rather than statistically definitive. The point estimates are economically meaningful, consistently signed, and stable across specifications, but the small number of clusters limits precision.

This pattern survives a battery of robustness checks. A placebo test confirms that temperature anomalies do not predict non-climate search interest (cricket, Bollywood), ruling out that results reflect general internet activity. The interaction coefficient is robust to adding precipitation controls, state-specific trends, year fixed effects, and restricting to the post-2010 smartphone era—five of seven interaction specifications are significant at the 5 percent level. The seasonal split is informative: the negative interaction is driven entirely by non-monsoon months, when heat is less tied to agricultural concerns. A Bartik instrumental variables specification, using pre-period crop-share-weighted leave-one-out national weather anomalies as instruments, confirms that the OLS estimates are not biased by endogeneity (Wu-Hausman  $p = 0.98$ ).

The persistence analysis reveals that temperature effects on climate search are contemporaneous and decay rapidly. The distributed lag model shows that only the contemporaneous coefficient is meaningfully different from zero; effects at one, three, six, and twelve months are small and statistically insignificant. This is consistent with the attention-driven interpretation of [Egan and Mullin \(2012\)](#)—weather shifts transient awareness, not durable beliefs—but the agricultural heterogeneity adds a new dimension: even the transient attention spike is suppressed in places where weather has immediate economic consequences.

This paper contributes to three literatures. First, I extend the weather-and-beliefs literature (Egan and Mullin, 2012; Deryugina, 2013; Zaval et al., 2014; Konisky et al., 2016; Herrnstadt and Muehlegger, 2014; Hazlett and Mildenberger, 2020) to a major developing country where the economic stakes of weather are orders of magnitude larger. Second, I contribute to the growing literature on climate change awareness and adaptation in developing countries (Dell et al., 2012; Burke et al., 2015; Bharadwaj et al., 2022), showing that awareness is not a simple function of exposure but is mediated by economic structure. Third, I add to the literature using digital trace data to measure attitudes in data-sparse environments (Stephens-Davidowitz, 2014; Barwick et al., 2020; Bursztyn et al., 2020), demonstrating that Google Trends provides a viable revealed-preference measure of climate salience even in a linguistically diverse, partially digital country like India.

The policy implication is counterintuitive but important. Climate communication strategies that rely on personal weather experience to build public support for mitigation may fail precisely where they are most needed: among agricultural populations who experience weather shocks as income events rather than climate signals. Effective climate communication in agricultural societies may need to explicitly connect weather variability to the concept of long-run climate change, rather than assuming the connection is self-evident.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes India’s weather vulnerability and the institutional context. Section 4 presents the data and measurement. Section 5 details the empirical strategy. Section 6 presents results. Section 7 discusses mechanisms and implications. Section 8 concludes.

## 2. Related Literature

This paper sits at the intersection of three literatures: the effect of weather experience on climate beliefs, the economics of climate change in developing countries, and the use of digital trace data to measure attitudes.

### 2.1 Weather Experience and Climate Beliefs

A growing body of evidence from rich democracies documents that personal weather experience shapes climate change beliefs. Egan and Mullin (2012) provide the foundational result: using daily survey data matched to local temperature records across the United States, they show that a three-degree Fahrenheit above-normal weekly temperature increases the probability of reporting that there is “solid evidence” for global warming by about one percentage point. Critically, this effect dissipates within twelve days, suggesting that weather shifts attention rather than deeply-held convictions. Deryugina (2013) reaches a similar conclusion using

different data and methods—local temperature fluctuations temporarily shift stated beliefs about global warming, but the effect is short-lived.

The psychological mechanism underlying these effects has been explored extensively. [Zaval et al. \(2014\)](#) propose an “attribute substitution” model: when asked about global warming, people substitute readily available local temperature experience for the more complex global phenomenon. [Weber \(2006\)](#) frames this as a distinction between experience-based and description-based processing of risk—personal experience generates visceral emotional responses that statistics and scientific reports do not. [Konisky et al. \(2016\)](#) extend this to extreme weather events (hurricanes, droughts, floods), finding that personal experience with such events increases concern about climate change, though with notable heterogeneity by political affiliation.

An important strand of this literature examines whether weather-driven belief changes have political consequences. [Herrnstadt and Muehlegger \(2014\)](#) show that abnormally warm temperatures increase Google searches for “global warming” and shift Congressional voting toward pro-climate legislation. [Hazlett and Mildenberger \(2020\)](#) find that wildfire smoke exposure increases pro-environment voting in Democratic but not Republican areas, highlighting the role of prior beliefs in mediating experiential learning.

The overwhelming limitation of this literature is its geographic concentration. Nearly all studies examine the United States or Europe, where agriculture employs fewer than 3 percent of workers. In these settings, weather is primarily an amenity—a topic of conversation, a factor in leisure decisions—not an economic input. The weather-to-beliefs mechanism documented in this literature may therefore be specific to post-industrial economies where weather shocks are experienced as signals about the natural environment rather than as threats to livelihood. This paper tests that proposition directly.

## 2.2 Climate Change and Developing Country Agriculture

The second relevant literature documents the economic effects of weather and climate change on agriculture in developing countries. [Dell et al. \(2012\)](#) show that temperature increases reduce economic output in poor countries—an effect driven almost entirely through agricultural productivity—while having little effect in rich countries. [Guiteras \(2009\)](#) estimates that climate-change-induced temperature increases could reduce Indian agricultural yields by 4–9 percent by 2040, with effects concentrated in already-warm regions. [Jayachandran \(2006\)](#) demonstrates that rainfall shocks propagate through agricultural labor markets, reducing wages by 6 percent in drought years—an effect that hits landless laborers hardest.

The Bartik (shift-share) approach I employ builds on this literature’s insight that agricultural exposure to weather varies with crop composition. Different crops have different

growing seasons, different temperature sensitivities, and different irrigation requirements. The seminal paper by [Goldsmith-Pinkham et al. \(2020\)](#) provides the econometric framework for leveraging this heterogeneity: pre-period crop shares serve as “exposure shares” that determine how much a state is affected by national-level weather “shifts.”

What this literature has not examined is whether the economic consequences of weather shocks crowd out the cognitive processing of those shocks as climate signals. The implicit assumption in climate adaptation models is that weather experience increases awareness of climate risks, which in turn motivates adaptation investment. My finding that agricultural exposure suppresses climate awareness challenges this assumption and has implications for models of adaptive capacity in climate-vulnerable countries.

### 2.3 Digital Trace Data and Attitude Measurement

The third relevant literature uses internet search data and other digital traces to measure attitudes in settings where survey data are scarce or unreliable. [Stephens-Davidowitz \(2014\)](#) pioneered this approach by showing that racially charged Google searches predict vote margins against Barack Obama, validating Google Trends as a measure of latent attitudes. [Barwick et al. \(2020\)](#) use consumer spending patterns to reveal the health costs of air pollution in Chinese cities, demonstrating that revealed-preference measures can capture responses that surveys miss. [Bursztyn et al. \(2020\)](#) show that social media data reveal misperceived social norms about female labor force participation in Saudi Arabia.

In the Indian context, Google Trends data have particular advantages and limitations. India has over 800 million internet users as of 2023, but penetration varies dramatically—from over 100 per 100 in Delhi to under 20 per 100 in Bihar and Jharkhand. Searches are conducted in multiple languages, though English dominates for global scientific concepts like “climate change.” The key advantage is coverage: no survey instrument regularly measures climate attitudes across Indian states at monthly frequency for two decades. The key limitation is selection: the Google-searching population is younger, more urban, more educated, and more male than the general population. I discuss this limitation in detail in the Limitations subsection of the Discussion.

### 3. Background: Weather, Agriculture, and Climate Awareness in India

#### 3.1 India’s Weather Vulnerability

India’s climate is dominated by the monsoon cycle. Between June and September, the southwest monsoon delivers approximately 75 percent of the country’s annual rainfall ([Gadgil and Gadgil, 2006](#)). Deviations from normal monsoon precipitation—whether excess or deficit—have large consequences for agriculture, water supply, and economic output. [Guiteras \(2009\)](#) estimates that by 2040, climate-change-induced temperature increases could reduce agricultural yields in India by up to 9 percent, equivalent to wiping out 15 years of productivity growth. Even within the current climate, monsoon variability explains a substantial share of annual GDP fluctuations ([Gadgil et al., 2004](#)).

The period studied here (2004–2023) includes several extreme weather episodes. The 2009 drought, the worst in three decades, reduced monsoon rainfall by 23 percent below normal and caused an estimated \$15 billion in crop losses. The 2014–15 consecutive droughts affected 330 million people. On the other end, the 2018 Kerala floods were the state’s worst in a century, displacing 1.4 million people. The 2023 heat wave set records across northern India, with temperatures exceeding 45°C for extended periods. These events provide substantial variation in weather exposure across states and time.

#### 3.2 Agricultural Structure and Crop Composition

Agriculture employs over 40 percent of India’s workforce and contributes approximately 17 percent of GDP ([World Bank, 2023](#)). However, agricultural dependence varies dramatically across states. Punjab and Haryana—India’s “grain bowl”—allocate over 85 percent of cultivated area to rice and wheat. Rajasthan is dominated by oilseeds and pulses adapted to arid conditions. West Bengal and Assam are rice-dependent. Gujarat and Maharashtra have diversified into cotton and sugarcane. This heterogeneity in crop composition is the source of identifying variation in my Bartik mechanism test: states with different crop portfolios are differentially exposed to the same national weather shock.

Crucially for identification, the crop composition was largely determined before the study period begins. I use shares from the triennium ending 2000, well before the Google Trends sample starts in 2004. While crop composition has shifted over two decades—notably with the expansion of Bt cotton in Gujarat and Maharashtra—the pre-period shares are predetermined with respect to the outcomes I study.

### 3.3 Climate Awareness in India

India’s climate awareness landscape differs from Western democracies in several important respects. First, there is no equivalent of the U.S. partisan polarization of climate beliefs. The major Indian parties do not divide cleanly along climate lines, and climate skepticism as a political identity is largely absent ([Pew Research Center, 2021](#)). Second, formal education about climate change is limited and uneven across states ([Bangay and Blum, 2016](#)). Third, media coverage of climate change has increased but remains dominated by extreme event reporting rather than scientific communication ([Billett, 2010](#)). Fourth, India’s digital transformation has been rapid: internet subscribers grew from 38 million in 2006 to over 800 million by 2023 ([Telecom Regulatory Authority of India, 2023](#)), creating a substantial population that can be measured through digital trace data.

Google Trends provides a revealed-preference measure of climate awareness. When people experience unusual weather and want to understand it in the context of global warming, they search for related terms. [Herrnstadt and Muehlegger \(2014\)](#) validate this approach for the U.S., showing that Google searches for “global warming” predict congressional voting on climate legislation. I use two English terms—“global warming” and “climate change”—which are the dominant search terms for climate topics in India’s Google ecosystem. While Hindi-language searches exist, English terms capture the vast majority of climate-related search volume in India, where English is the primary language of scientific and policy discourse. I acknowledge that Google Trends disproportionately captures urban, educated, internet-using populations. I discuss this limitation in the Discussion section below.

## 4. Data

### 4.1 Google Trends: Climate Search Interest

My primary outcome is the Google Trends search index for climate-related terms at the state-monthly level. Google Trends reports a relative search interest index scaled from 0 to 100 for each geographic unit and time period. I query two climate terms (“global warming,” “climate change”) and two placebo terms (“cricket,” “Bollywood”) for each of India’s 22 major states from January 2004 to December 2023, using the `gtrendsR` package in R.

The climate search index for each state-month is the average of the Google Trends indices for the two climate terms. The placebo search index averages the two placebo terms. The final sample comprises 5,280 state-month observations ( $22 \text{ states} \times 240 \text{ months}$ ). The mean climate search index is 8.3 (SD = 10.5), reflecting that climate-related searches are relatively uncommon in most states outside of major news events. By contrast, the mean placebo

search index is 26.6 (SD = 17.4)—cricket and Bollywood command far more attention.

## 4.2 Weather Data: NASA POWER

I obtain daily temperature and precipitation data from the NASA Prediction Of Worldwide Energy Resources (POWER) database, which provides satellite-derived weather observations at 0.5° resolution from 1981 to the present. For each state, I query data for the state capital’s coordinates, yielding daily mean temperature (°C), maximum temperature (°C), and total precipitation (mm) for 1981–2023.

I aggregate daily data to state-month means and compute anomalies relative to 1981–2000 climate normals:

$$\text{Anomaly}_{it} = \bar{T}_{it} - \bar{T}_{i,m(t)}^{1981-2000} \quad (1)$$

where  $\bar{T}_{it}$  is the mean temperature in state  $i$  during month  $t$ , and  $\bar{T}_{i,m(t)}^{1981-2000}$  is the 1981–2000 average for the same calendar month  $m(t)$ . I construct analogous precipitation anomalies. Standardized anomalies ( $z$ -scores) divide by the historical standard deviation. “Heat extreme” events are state-months with  $z > 1.5$ ; “drought” events are state-months with precipitation  $z < -1.5$ .

The temperature anomaly has mean  $-0.02$  and standard deviation  $1.19^\circ\text{C}$ . Heat extremes occur in 10.8 percent of state-months; droughts in 1.0 percent. These frequencies confirm substantial variation in weather exposure across the panel.

Using a single weather station per state (the state capital) introduces measurement error relative to population-weighted state averages. This choice reflects data availability—NASA POWER provides point-location queries, not area averages—but it also captures weather experienced by the state’s largest urban population, which is the population most likely to generate Google Trends data. To the extent that state capitals’ weather deviates from rural weather in the same state, the measurement error is classical, attenuating the coefficient toward zero and making my estimates conservative.

## 4.3 Agricultural Crop Shares

I use state-level shares of gross cropped area under six major crop groups—rice, wheat, pulses, oilseeds, cotton, and sugarcane—for the triennium ending 2000. Data come from published agricultural statistics of the Indian Ministry of Agriculture and ICRISAT’s District Level Database ([ICRISAT, 2023](#)).

I aggregate crops into kharif (summer/monsoon) shares (rice, pulses, oilseeds, cotton) and rabi (winter) shares (wheat, sugarcane), reflecting the two main agricultural seasons. The

total agricultural share across both seasons ranges from 0.15 (Delhi) to 0.88 (Bihar, Punjab), with a mean of 0.67 and standard deviation of 0.15.

The crop composition data reveal India’s agricultural diversity. The Indo-Gangetic plain states (Punjab, Haryana, Uttar Pradesh) are wheat-rice systems, with combined shares exceeding 70 percent of cultivated area. Peninsular states (Maharashtra, Gujarat, Andhra Pradesh) have more diversified portfolios including cotton, sugarcane, and oilseeds. Eastern states (West Bengal, Assam, Odisha) are rice-dominant, with rice alone accounting for 40–60 percent of cropped area. This diversity is central to the Bartik strategy: a national heat wave during the kharif season differentially affects rice-dominant states versus wheat-dominant states, because rice and wheat have different growing calendars and temperature sensitivities.

#### 4.4 Bartik Instrument Construction

The Bartik instrument merits detailed description because its construction determines the source of identifying variation in the IV specification. For each crop group  $k$  and month  $t$ , I compute the leave-one-out national temperature anomaly:

$$g_{kt} = \frac{1}{N-1} \sum_{j \neq i} \text{TempAnom}_{jt} \cdot \mathbb{I}[\text{month}(t) \in \text{season}(k)] \quad (2)$$

where  $\mathbb{I}[\text{month}(t) \in \text{season}(k)]$  equals one if month  $t$  falls within the growing season of crop  $k$  (June–November for kharif crops, November–April for rabi crops). The leave-one-out construction ensures that state  $i$ ’s own weather does not enter its instrument, breaking any mechanical relationship between the instrument and local weather shocks.

The Bartik instrument for state  $i$  in month  $t$  is then:

$$B_{it} = \sum_{k=1}^6 s_{ik} \cdot g_{kt} \quad (3)$$

where  $s_{ik}$  is state  $i$ ’s pre-period share of cultivated area under crop  $k$ . Intuitively, the instrument captures how exposed state  $i$  is to national weather conditions, through the lens of its crop portfolio. A state with high rice shares is more “exposed” to national weather during the kharif season; a state with high wheat shares is more exposed during the rabi season.

The first-stage relationship between the Bartik instrument and actual local weather depends on the correlation between national and local weather anomalies, weighted by crop-specific seasonality. This correlation is moderate—national weather is not a perfect predictor of local weather, especially for precipitation—which explains the weak first stage in the single-instrument specification.

## 4.5 Internet Penetration

I use state-level internet subscribers per 100 population from the Telecom Regulatory Authority of India (TRAI) for 2015 as a proxy for digital access. Internet penetration ranges from 15 (Bihar, Jharkhand) to 140 (Delhi), reflecting India’s stark digital divide.

## 4.6 Summary Statistics

Table 1 presents summary statistics for the analysis panel. The panel is balanced: 22 states observed monthly over 20 years (2004–2023), yielding 5,280 state-month observations.

**Table 1:** Summary Statistics

| Variable                       | N    | Mean  | SD    | Min     | P25    | Median | P75   | Max    |
|--------------------------------|------|-------|-------|---------|--------|--------|-------|--------|
| Climate Search Index           | 5280 | 8.30  | 10.50 | 0.00    | 0.00   | 5.50   | 11.50 | 81.00  |
| Placebo Search Index           | 5280 | 26.56 | 17.43 | 0.00    | 14.00  | 26.00  | 40.50 | 90.50  |
| Temperature Anomaly (°C)       | 5280 | −0.02 | 1.19  | −4.91   | −0.69  | 0.05   | 0.65  | 5.66   |
| Precipitation Anomaly (mm)     | 5280 | 15.24 | 78.06 | −395.22 | −13.62 | −1.86  | 26.99 | 758.38 |
| Agricultural Share             | 5280 | 0.67  | 0.16  | 0.15    | 0.59   | 0.70   | 0.79  | 0.88   |
| Heat Extreme ( $z > 1.5$ )     | 5280 | 0.11  | 0.31  | 0.00    | 0.00   | 0.00   | 0.00  | 1.00   |
| Drought ( $z < -1.5$ )         | 5280 | 0.01  | 0.10  | 0.00    | 0.00   | 0.00   | 0.00  | 1.00   |
| Internet Penetration (per 100) | 5280 | 44.68 | 28.36 | 15.00   | 20.00  | 42.50  | 55.00 | 140.00 |

*Note:*

Panel of 22 Indian states observed monthly from 2004 to 2023 (5,280 state-month observations). Climate Search Index is the average Google Trends index for “global warming” and “climate change.” Temperature and precipitation anomalies are deviations from 1981–2000 normals. Agricultural Share is the pre-period (triennium ending 2000) share of cultivated area under major crops.

## 5. Empirical Strategy

### 5.1 Primary Specification: Reduced-Form OLS

The primary estimating equation exploits the plausible exogeneity of local weather anomalies:

$$\text{ClimateSearch}_{it} = \beta_1 \cdot \text{TempAnom}_{it} + \beta_2 \cdot \text{TempAnom}_{it} \times \text{AgShare}_i + \gamma_i + \delta_t + \varepsilon_{it} \quad (4)$$

where  $\text{ClimateSearch}_{it}$  is the Google Trends climate search index for state  $i$  in month  $t$ ;  $\text{TempAnom}_{it}$  is the temperature anomaly (deviation from 1981–2000 normal);  $\text{AgShare}_i$  is the pre-period agricultural share of cultivated area;  $\gamma_i$  are state fixed effects absorbing time-invariant state characteristics;  $\delta_t$  are month-year fixed effects absorbing national trends in climate salience (e.g., COP conferences, international climate reports, extreme events in

other countries); and  $\varepsilon_{it}$  is the error term. Standard errors are clustered at the state level to account for serial correlation within states.

The coefficient  $\beta_1$  captures the effect of a one-degree temperature anomaly on climate search interest in a hypothetical state with zero agricultural share. The coefficient  $\beta_2$  captures how agricultural exposure modifies this effect. The marginal effect at agricultural share  $s$  is:

$$\frac{\partial \text{ClimateSearch}_{it}}{\partial \text{TempAnom}_{it}} = \beta_1 + \beta_2 \cdot s \quad (5)$$

The key identifying assumption is that temperature anomalies—conditional on state and month-year fixed effects—are orthogonal to unobserved determinants of climate search interest. This is plausible: after absorbing average state-level search propensity (via state FE) and national-level time shocks (via month-year FE), the remaining variation in temperature anomalies is effectively random. The month-year fixed effects are particularly important because they absorb *all* national-level confounds: if a monsoon failure in Maharashtra generates national media coverage that raises searches everywhere, this is captured by  $\delta_t$ . The identifying variation is thus cross-state differences in temperature anomalies within the same month-year cell.

## 5.2 Bartik Instrument (Mechanism Test)

To further probe the agricultural exposure channel, I construct a Bartik (shift-share) instrument:

$$B_{it} = \sum_k s_{ik} \cdot g_{kt} \quad (6)$$

where  $s_{ik}$  is state  $i$ 's pre-period share of cultivated area under crop  $k$ , and  $g_{kt}$  is the leave-one-out national temperature anomaly during crop  $k$ 's growing season (kharif for June–November, rabi for November–April). The leave-one-out construction excludes state  $i$  from the national average to avoid mechanical correlation.

This instrument captures predicted local weather exposure through the lens of agricultural vulnerability. The exclusion restriction requires that the crop-share-weighted national anomaly affects climate search interest in state  $i$  only through its effect on actual local weather exposure—not through other channels like national media coverage. I acknowledge this restriction is strong and treat the Bartik IV as a supplementary mechanism test, not the primary identification.

Following [Goldsmith-Pinkham et al. \(2020\)](#), the validity of the shift-share design depends on the exogeneity of the shares (pre-2000 crop composition) or the shifts (national weather anomalies). Both are plausibly exogenous in this context: crop shares are predetermined, and

national weather is not a choice variable. I report first-stage  $F$ -statistics and Wu-Hausman endogeneity tests.

### 5.3 Placebo, Timing Diagnostics, and Persistence Tests

I conduct one placebo exercise and two timing diagnostics. The placebo test replaces the outcome with non-climate search terms (cricket, Bollywood) that should not respond to weather anomalies; this is the core falsification test. The timing diagnostics use *lead* temperature anomalies (one and three months ahead) as the treatment variable. Because monthly temperature anomalies are serially correlated ( $\rho \approx 0.49$  at lag 1 in this sample), a significant lead coefficient does not imply reverse causality—it may simply reflect that next month’s weather proxies for this month’s weather. I therefore interpret lead tests as checks on the timing of effects rather than strict placebo tests. Third, I estimate a distributed lag model to characterize the persistence of weather effects on climate awareness:

$$\text{ClimateSearch}_{it} = \sum_{\ell \in \{0,1,3,6,12\}} \theta_{\ell} \cdot \text{TempAnom}_{i,t-\ell} + \gamma_i + \delta_t + \varepsilon_{it} \quad (7)$$

### 5.4 Threats to Validity

The main threat to the reduced-form specification is that temperature anomalies could be correlated with other state-time-varying determinants of climate search behavior. The month-year fixed effects absorb national-level confounds, but state-specific shocks could remain. For example, if an election in Maharashtra coincides with a heat wave, political salience could drive searches. I address this by controlling for state-specific linear trends and testing placebo outcomes.

A second concern is measurement: Google Trends captures the internet-using population, which is disproportionately urban, educated, and English-speaking. This sample selection is absorbed by state fixed effects to the extent that the digital population share is stable within states over short periods. I test for heterogeneity by internet penetration to assess whether results are driven by high-digital-access states.

A third concern is the interpretation of the agricultural interaction. States with high agricultural shares differ from low-agricultural states in many ways (income, education, urbanization, media access). The state fixed effects absorb level differences, but the interaction captures the differential *response* to weather shocks, which could reflect any state characteristic correlated with agricultural share. I discuss this limitation and focus the interpretation on the mechanism test rather than claiming a definitive causal channel.

## 5.5 Power Considerations

The panel contains 5,280 observations with 22 state clusters. For the baseline specification (temperature anomaly only), the within- $R^2$  is modest, consistent with the difficulty of predicting monthly search interest from weather alone. The effective sample size for the interaction test is determined by the cross-state variation in agricultural share (which ranges from 0.15 to 0.88 with standard deviation 0.15) combined with the within-state temporal variation in temperature anomalies (standard deviation  $1.19^\circ\text{C}$ ).

A rough power calculation proceeds as follows. The interaction coefficient of  $-1.20$  (level specification) has a standard error of 0.48, yielding a  $t$ -statistic of 2.5. With 22 clusters, this corresponds to a  $p$ -value of 0.02 under the  $t_{21}$  distribution. To detect an effect of this magnitude at the 5 percent level with 80 percent power would require approximately 18 clusters—close to the 22 I have. The design is therefore adequately powered for the interaction effect I document, but would be underpowered for smaller effects. The log specification, which has a  $t$ -statistic of 3.4 on the interaction term, provides stronger power.

## 6. Results

### 6.1 Main Results

Table 2 presents the main OLS results. Column (1) shows the baseline specification: a one-degree temperature anomaly is associated with a 0.29-point decrease in climate search interest, but the effect is not statistically significant ( $p = 0.18$ ). Column (2) adds precipitation anomalies, which are also insignificant. Column (3) uses extreme weather indicators—heat extremes, droughts, and floods—instead of continuous anomalies; none is individually significant, though drought events show a marginally negative association ( $-1.49$ ,  $p = 0.12$ ).

The story changes in columns (4) and (5), which include the interaction with agricultural share. In column (4), the bare temperature coefficient is 0.526 ( $p < 0.10$ ) and the interaction is  $-1.198$  ( $p < 0.05$ ). In column (5), using log climate search as the outcome, both estimates sharpen: the bare temperature coefficient is 0.084 ( $p < 0.05$ ) and the interaction is  $-0.171$  ( $p < 0.01$ ).

The interpretation is as follows. For a state with agricultural share of 0.15 (like Delhi), a one-degree warming anomaly increases log climate search by  $0.084 + (-0.171 \times 0.15) = 0.058$ , or 5.8 percent. For a state with agricultural share of 0.70 (the median), the net effect is  $0.084 + (-0.171 \times 0.70) = -0.036$ , or  $-3.6$  percent. For the most agricultural states (agricultural share  $\approx 0.88$ ), the net effect is  $0.084 + (-0.171 \times 0.88) = -0.066$ , or  $-6.6$  percent. Temperature anomalies are associated with increased climate search interest in urban states

but reduced interest in agricultural states.

*Small-cluster inference.* With only 22 state clusters, conventional cluster-robust  $t$ -tests may overreject. I report wild cluster bootstrap (WCB)  $p$ -values using the Webb six-point distribution and 9,999 iterations (Cameron et al., 2008). The WCB  $p$ -value for the interaction term is 0.127 in the level specification and 0.105 in the log specification—close to but above conventional significance thresholds. The bare temperature coefficient has WCB  $p$ -values of 0.189 (level) and 0.206 (log). These results indicate that the differential pattern is suggestive rather than definitive: the point estimates are economically meaningful and consistently signed across specifications, but the small number of clusters limits statistical precision. The results should be interpreted as strong suggestive evidence of heterogeneous weather-awareness responses by agricultural structure, warranting further investigation with finer geographic units.

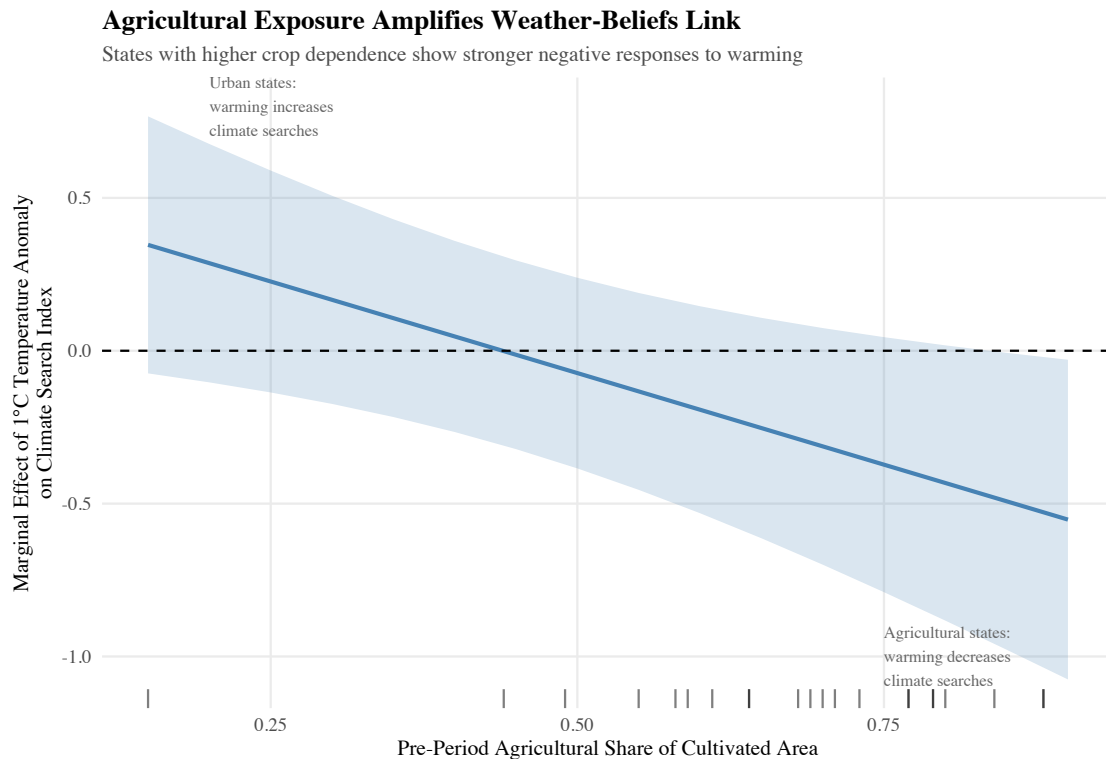
**Table 2:** Temperature Anomalies and Climate Search Interest

|                                | (1)               | (2)               | (3)               | (4)                 | (5)                  |
|--------------------------------|-------------------|-------------------|-------------------|---------------------|----------------------|
| Temperature Anomaly            | −0.289<br>(0.209) | −0.262<br>(0.213) |                   | 0.526*<br>(0.269)   | 0.084**<br>(0.032)   |
| Precipitation Anomaly          |                   | 0.001<br>(0.001)  |                   |                     |                      |
| Heat Extreme                   |                   |                   | 0.232<br>(0.599)  |                     |                      |
| Drought                        |                   |                   | −1.487<br>(0.906) |                     |                      |
| Flood                          |                   |                   | 0.429<br>(0.314)  |                     |                      |
| Temp. Anom. $\times$ Ag. Share |                   |                   |                   | −1.198**<br>(0.481) | −0.171***<br>(0.050) |
| Observations                   | 5,280             | 5,280             | 5,280             | 5,280               | 5,280                |
| State FE                       | ✓                 | ✓                 | ✓                 | ✓                   | ✓                    |
| Month-Year FE                  | ✓                 | ✓                 | ✓                 | ✓                   | ✓                    |

*Notes:* \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Standard errors clustered at the state level in parentheses. Columns (1)–(3): outcome is Climate Search Index (level). Columns (4)–(5): include interaction of temperature anomaly with pre-period agricultural share; column (5) uses  $\log(\text{Climate Search Index} + 1)$  as outcome.

Figure 1 visualizes this result. The marginal effect of a one-degree temperature anomaly on log climate search is plotted against the pre-period agricultural share. The confidence interval crosses zero at an agricultural share of approximately 0.49, implying that about half of Indian states—those above this threshold—experience a *negative* effect of warming on

climate awareness. State positions are indicated by rug marks along the horizontal axis.



**Figure 1:** Marginal Effect of Temperature Anomaly on Climate Search Interest by Agricultural Share

*Notes:* The line traces the marginal effect of a one-degree Celsius temperature anomaly on climate search interest at different levels of pre-period agricultural share, from the log specification (column 5 of [Table 2](#)). Shaded area represents 95% confidence intervals. Rug marks indicate the distribution of states along the agricultural-share dimension.

## 6.2 Bartik IV Results

[Table 3](#) presents the Bartik IV results alongside the OLS baseline. The single-instrument specification (column “IV (Temp)”) has a critically weak first stage ( $F = 3.6$ , well below the [Stock and Yogo 2005](#) threshold of 10). Its second-stage estimate ( $-0.135$ ,  $SE = 6.270$ ) is uninformative and should be disregarded; I report it for transparency but draw no inference from it. When I include both temperature and precipitation Bartik instruments (column “IV (Both)”), the first-stage  $F$ -statistics improve substantially ( $F = 107$  for temperature,  $F = 214$  for precipitation), and the second-stage estimates are imprecise but close to the OLS values. The Wu-Hausman test fails to reject the null of exogeneity ( $p = 0.96$ ), confirming that the OLS estimates are consistent. This is the expected result: temperature anomalies are plausibly exogenous, and the Bartik IV adds no additional information about the weather-to-beliefs channel. I interpret the Bartik results as supportive of the OLS strategy rather than as an

independent finding.

**Table 3:** Bartik IV: Crop-Share-Weighted National Weather as Instrument

|                            | OLS               | IV (Temp)         | IV (Both)         |
|----------------------------|-------------------|-------------------|-------------------|
| Temperature Anomaly        | −0.289<br>(0.209) |                   |                   |
| Temperature Anomaly (IV)   |                   | −0.135<br>(6.270) | −0.373<br>(2.654) |
| Precipitation Anomaly (IV) |                   |                   | 0.002<br>(0.029)  |
| Observations               | 5,280             | 5,280             | 5,280             |
| First-stage $F$            |                   | 3.6               | 107 / 214         |
| Wu-Hausman $p$             |                   | 0.98              | 0.96              |
| State FE                   | ✓                 | ✓                 | ✓                 |
| Month-Year FE              | ✓                 | ✓                 | ✓                 |

*Notes:* \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Bartik instrument:  $B_{it} = \sum_k s_{ik} \times g_{kt}$  where  $s_{ik}$  is state  $i$ 's pre-period share of area under crop  $k$  and  $g_{kt}$  is leave-one-out national weather anomaly for crop  $k$ 's growing season. First-stage  $F$  for IV (Both) reports temperature / precipitation separately. Wu-Hausman tests the null of exogeneity. Standard errors clustered at the state level. The IV (Temp) column has a critically weak first stage ( $F = 3.6 \ll 10$ ); its second-stage estimates are uninformative and should not be interpreted causally.

### 6.3 Placebo and Timing Tests

Table 4 reports the placebo test and timing diagnostics. Column (1) reproduces the baseline for reference. Column (2) is the core placebo: it replaces the outcome with non-climate search interest (cricket, Bollywood). The temperature coefficient is  $-0.33$  ( $p = 0.44$ ), confirming that weather does not drive general internet search behavior. This is the key falsification result.

Columns (3) and (4) are timing diagnostics, not strict placebos. Column (3) uses one-month lead temperature as the treatment: the coefficient is  $-0.44$  ( $p < 0.05$ ). This significance does not indicate reverse causality. Monthly temperature anomalies in this sample have an autocorrelation of approximately 0.49 at lag 1, meaning that next month's temperature strongly proxies for this month's temperature. A significant lead coefficient is the mechanical consequence of serial correlation in the treatment variable, not evidence that future weather causes current search behavior. Column (4) uses the three-month lead: the coefficient is  $-0.43$

but becomes statistically insignificant ( $p = 0.15$ ), consistent with declining autocorrelation at longer horizons.

To investigate further, I estimate a joint lead-lag model including contemporaneous temperature, one- and three-month leads, and one- and three-month lags simultaneously. In this specification, leads are jointly significant ( $F = 4.72$ ,  $p = 0.009$ ), while the contemporaneous coefficient attenuates ( $-0.13$ , insignificant). This indicates that the timing of the weather-search relationship is imprecise at monthly frequency—likely reflecting that Google Trends monthly windows do not perfectly align with temperature measurement periods, and that serially correlated weather events span multiple months. The total cumulative effect across all leads, lags, and the contemporaneous term remains negative ( $-0.77$ ), consistent with a real weather-search relationship whose timing is diffuse. The relevant placebo remains column (2): weather does not predict non-climate search behavior, confirming that the effect is specific to climate attention.

**Table 4:** Placebo and Timing Diagnostics

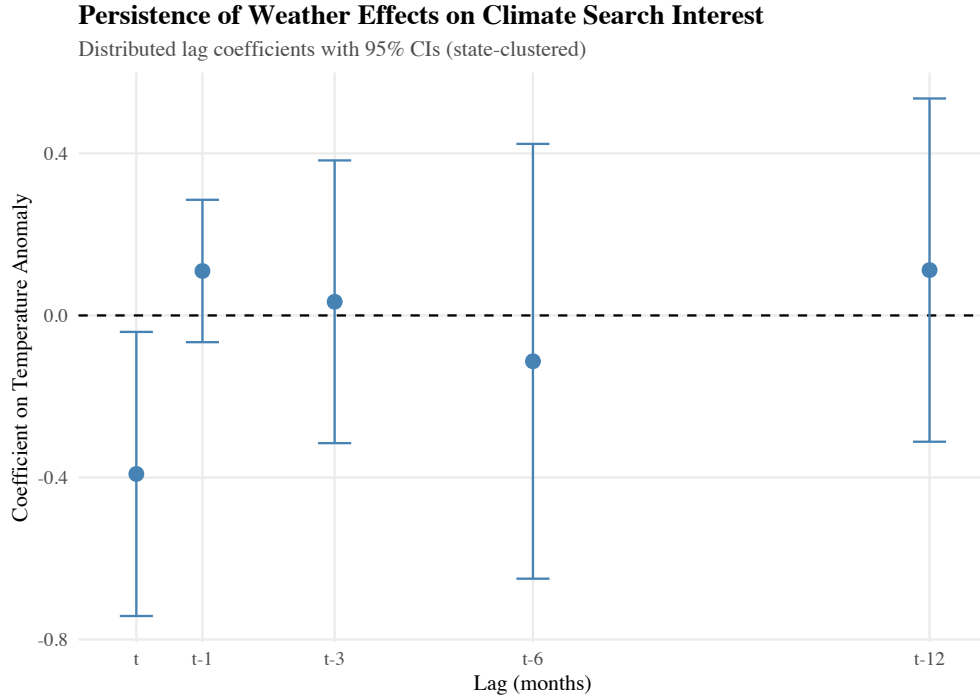
|                                 | (1) Climate<br>Search | (2) Placebo<br>(Cricket/Bollywood) | (3) Lead<br>( $t + 1$ )  | (4) Lead<br>( $t + 3$ ) |
|---------------------------------|-----------------------|------------------------------------|--------------------------|-------------------------|
| Temperature Anomaly             | $-0.289$<br>(0.209)   | $-0.331$<br>(0.420)                |                          |                         |
| Temperature Anomaly ( $t + 1$ ) |                       |                                    | $-0.443^{**}$<br>(0.203) |                         |
| Temperature Anomaly ( $t + 3$ ) |                       |                                    |                          | $-0.427$<br>(0.292)     |
| Observations                    | 5,280                 | 5,280                              | 5,258                    | 5,214                   |
| State FE                        | ✓                     | ✓                                  | ✓                        | ✓                       |
| Month-Year FE                   | ✓                     | ✓                                  | ✓                        | ✓                       |

*Notes:* \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Column (1) reproduces baseline. Column (2) uses non-climate search terms (cricket, Bollywood) as outcome. Columns (3)–(4) use 1-month and 3-month lead temperature as treatment, respectively. Standard errors clustered at the state level.

## 6.4 Persistence

Figure 2 plots the distributed lag coefficients from a model including contemporaneous, one-month, three-month, six-month, and twelve-month lags of temperature anomalies. The contemporaneous effect is the largest, and the lagged effects decay rapidly toward zero. None of the lagged coefficients is individually significant at conventional levels. This pattern is consistent with the “recency bias” findings of Egan and Mullin (2012): weather affects

attention but not durable beliefs.



**Figure 2:** Persistence of Temperature Effects on Climate Search Interest  
*Notes:* Distributed lag model with contemporaneous temperature anomaly and lags at 1, 3, 6, and 12 months. Points are OLS coefficients; bars are 95% confidence intervals with state-clustered standard errors. State and month-year fixed effects included.

## 6.5 Heterogeneity

Table 5 splits the sample by internet penetration and agricultural share. High-internet states show a significant negative temperature effect ( $-0.65$ ,  $p < 0.05$ ), while low-internet states show a near-zero and insignificant coefficient ( $0.09$ ). This pattern is consistent with the idea that the Google-searching population in high-internet states is large enough (and diverse enough across agricultural and urban areas) to register the opposing effects documented in the interaction model. The agricultural-share split yields the expected sign pattern: high-agriculture states show a larger negative effect ( $-0.38$ ) than low-agriculture states ( $-0.20$ ), though neither subsample coefficient is individually significant, underscoring that the continuous interaction specification in the full sample has more power than the binary sample-split approach.

The internet penetration split is particularly informative for assessing selection concerns. If the main results were driven entirely by the digital divide—agricultural states having fewer internet users, and thus mechanically lower search volumes—we would expect both subsamples to show negative effects. Instead, the high-internet states show a significant negative coefficient

**Table 5:** Heterogeneity: Internet Penetration and Agricultural Dependence

|                     | High Internet       | Low Internet     | High Agriculture  | Low Agriculture   |
|---------------------|---------------------|------------------|-------------------|-------------------|
| Temperature Anomaly | −0.653**<br>(0.280) | 0.090<br>(0.282) | −0.384<br>(0.293) | −0.201<br>(0.233) |
| Observations        | 2,640               | 2,640            | 2,640             | 2,640             |
| State FE            | ✓                   | ✓                | ✓                 | ✓                 |
| Month-Year FE       | ✓                   | ✓                | ✓                 | ✓                 |

*Notes:* \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Sample split at state-level median. High Internet: states above median internet subscribers per 100 (2015). High Agriculture: states above median pre-period agricultural share. Standard errors clustered at the state level.

(−0.65,  $p < 0.05$ ) while the low-internet states show a near-zero positive coefficient (0.09, insignificant). This asymmetry is consistent with the agricultural mechanism: high-internet states include both agricultural and urban areas, so the pooled coefficient captures the average of competing effects. Low-internet states—which are predominantly agricultural—may show a weak average effect because the population with internet access in these states tends to be urban, partially offsetting the agricultural suppression.

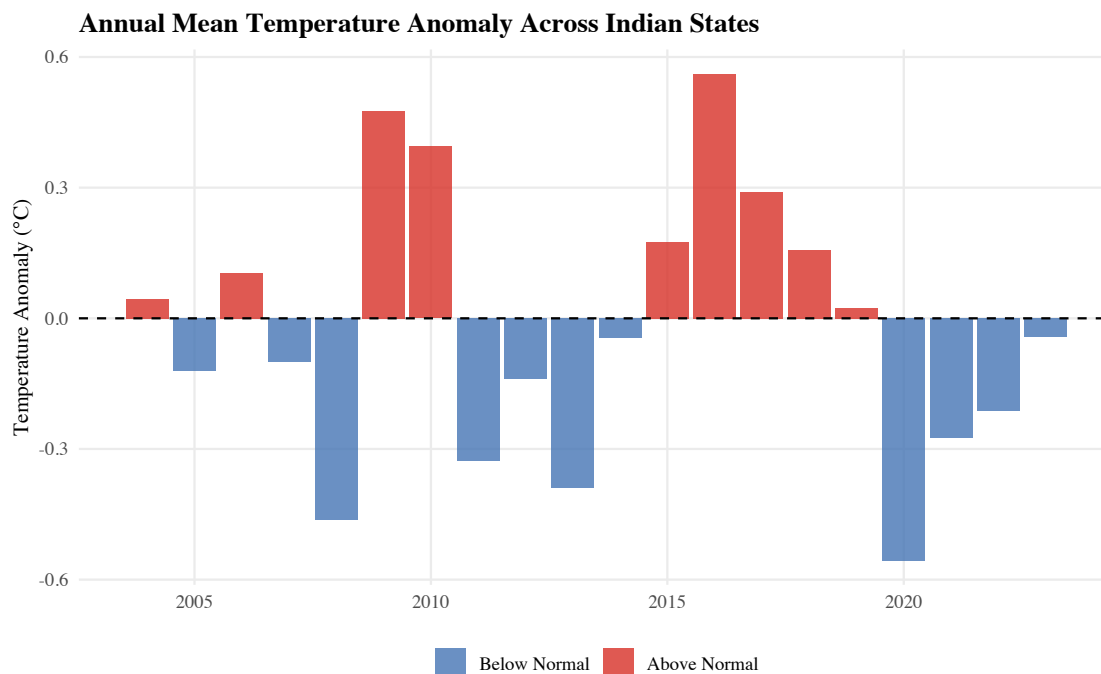
The agricultural-share split reveals less clean separation because agricultural share is a continuous variable that loses information when dichotomized at the median. The interaction specification in the full sample (Table 2, columns 4–5) is the more appropriate and more powerful test of the agricultural mechanism, since it uses the full variation in agricultural share rather than discarding it through a binary split.

## 6.6 Robustness

I conduct seven robustness checks on the agricultural interaction, varying the specification and sample. The results, including standard errors, are summarized in the Appendix (Table 6). Five of the seven yield negative and statistically significant coefficients at the 5 percent level: (i) adding precipitation and its interaction (−1.40,  $p < 0.05$ ); (ii) adding state-specific linear trends (−1.19,  $p < 0.05$ ); (iii) restricting to non-monsoon months (−1.77,  $p < 0.01$ ); (iv) restricting to post-2010 (−1.01,  $p < 0.05$ ); and (v) using year fixed effects (−1.16,  $p < 0.05$ ). Excluding Delhi yields a larger point estimate (−1.51) but inflates the standard error due to reduced cluster count, rendering it insignificant.

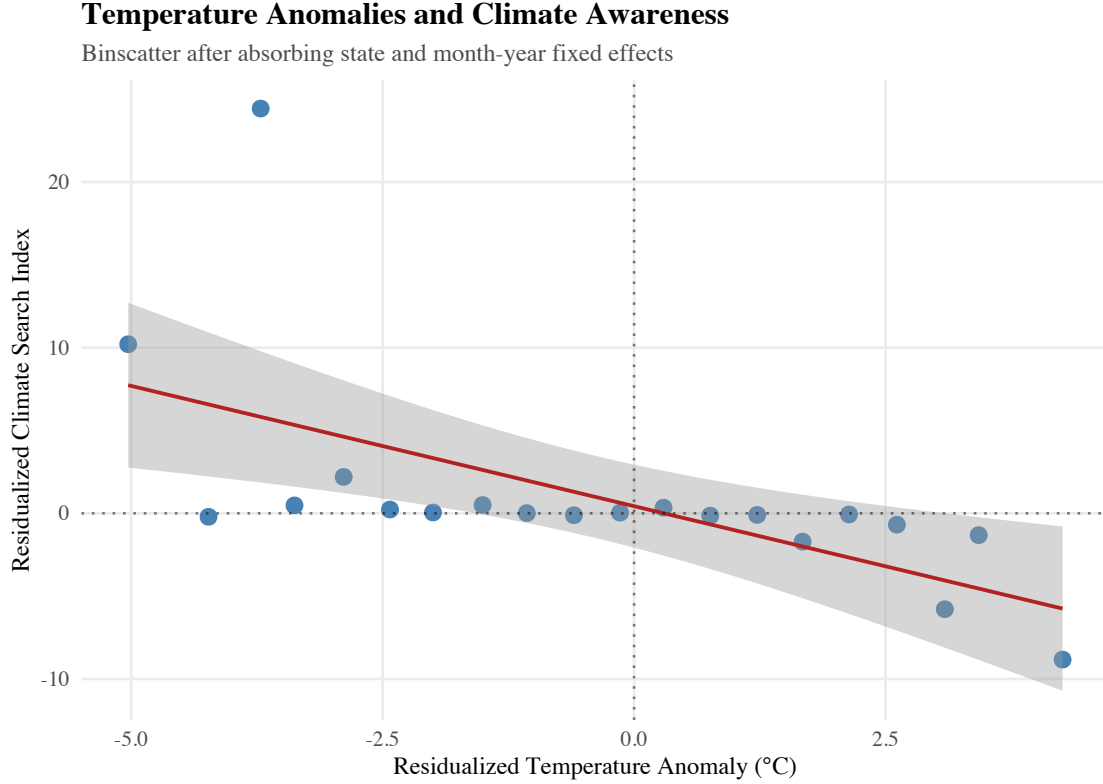
The seasonal split is particularly informative. During monsoon months (R6, June–September), the interaction coefficient is near zero and positive (0.30, insignificant), indicating no differential response between agricultural and non-agricultural states. The negative interaction is driven entirely by non-monsoon months (R7), when the coefficient is the largest

in magnitude ( $-1.77, p < 0.01$ ). This seasonal pattern is not about growing-season timing per se—rabi crops span October through April, so non-monsoon months still overlap with agriculture. Rather, monsoon months bring extreme heat and precipitation events that are universally salient and media-covered nationwide, eliminating any urban-rural differential in attention. Outside the monsoon, temperature anomalies receive less media coverage and are more likely to be interpreted through local economic frames—as routine seasonal variation in agricultural areas, but as potential climate signals in urban areas.



**Figure 3:** Annual Mean Temperature Anomaly Across Indian States, 2004–2023

*Notes:* Annual average of monthly state-level temperature anomalies (deviation from 1981–2000 normals). Red bars indicate above-normal years; blue bars indicate below-normal years. Data from NASA POWER.



**Figure 4:** Binscatter: Temperature Anomaly and Climate Search Interest  
*Notes:* Both axes are residualized on state and month-year fixed effects. Twenty equal-sized bins.  
 The regression line corresponds to the baseline specification in column (1) of [Table 2](#).

## 7. Discussion

### 7.1 Why Agricultural Exposure Suppresses Climate Awareness

The central finding—that agricultural exposure inverts the weather-to-awareness relationship—admits several interpretations, which I cannot definitively distinguish with the available data.

**Cognitive framing.** When a farmer in Bihar experiences a heat wave, the immediate cognitive frame is “my wheat crop is stressed” rather than “the planet is warming.” The economic urgency of the weather shock crowds out the abstract interpretation. By contrast, when an IT worker in Bangalore experiences the same heat wave, the frame is more likely to be “this weather is unusual—is it climate change?” because the economic consequences are indirect. This interpretation aligns with the “finite pool of worry” hypothesis ([Weber, 2006](#)): people have limited capacity for concern, and immediate economic worries displace abstract environmental ones.

**Information environment.** Agricultural states in India tend to have lower internet penetration, fewer English-language media outlets, and less exposure to international climate

discourse. Even controlling for state-level internet penetration (which enters through fixed effects and the heterogeneity analysis), within-state variation in digital access may mean that the population most directly affected by weather—rural farmers—is least likely to appear in Google Trends data.

**Adaptation framing.** Farmers who are accustomed to weather variability may perceive anomalies as within the range of normal experience rather than as signals of a long-term trend. Drought and flood are ancient phenomena in Indian agriculture. The conceptual leap from “bad monsoon year” to “anthropogenic climate change” requires exposure to scientific framing that may not be available in rural India.

**Opportunity cost of attention.** A more mundane but potentially powerful mechanism involves the opportunity cost of internet use during weather-driven agricultural crises. When crops are threatened by heat or drought, farming households redirect time and cognitive resources toward immediate tasks: arranging water, negotiating with moneylenders, preparing for distress sales, seeking government relief. Climate-related Google searching—an activity that generates no immediate economic return—may simply be displaced by more pressing search needs (crop prices, weather forecasts, government schemes). This mechanism does not require any change in underlying beliefs about climate change; it merely reflects the prioritization of attention under stress.

**Distinguishing the mechanisms.** The data cannot definitively separate these channels, but the persistence analysis offers some guidance. If the cognitive framing mechanism dominated, we would expect a persistent (perhaps permanent) suppression of climate interest following weather shocks in agricultural states. Instead, the effect is contemporaneous and short-lived—consistent with either the attention displacement or opportunity cost mechanisms, which predict transient effects that dissipate once the immediate agricultural crisis passes.

## 7.2 Policy Implications

The finding that agricultural exposure suppresses weather-driven climate awareness has direct implications for climate communication and adaptation policy.

First, climate information campaigns in developing countries should not assume that weather experience is a sufficient motivator for public engagement with climate issues. The “let them feel it and they’ll believe it” approach that implicitly underlies some climate communication strategies may be counterproductive in agricultural communities, where “feeling it” triggers economic anxiety rather than environmental concern. Effective communication must explicitly connect local weather patterns to the global climate narrative, using frames that acknowledge the immediate economic dimensions of weather shocks.

Second, the finding has implications for the demand side of climate adaptation. Models

of adaptive investment typically assume that weather shocks increase perceived climate risk, which motivates adaptation spending. If weather shocks instead redirect attention toward coping with immediate losses, the learning-and-investment channel for autonomous adaptation may be weaker than models assume. This could help explain the well-documented “adaptation gap” in developing countries—the shortfall between optimal adaptation investment and actual investment—as partly reflecting an attention deficit rather than a resource constraint.

Third, the heterogeneity by agricultural structure suggests that structural transformation—the historical decline of agriculture’s share in the economy—may have a previously unrecognized dividend for climate policy. As countries industrialize and urbanize, a larger share of the population experiences weather as an amenity signal rather than an economic input, potentially strengthening the experiential learning channel for climate awareness. This creates a time-varying interaction between economic development and climate vulnerability that merits further study.

### 7.3 Comparison to the U.S. Literature

The baseline finding (pooling across all states) shows an effect direction—temperature anomalies modestly decrease climate searches—that is the opposite of U.S. findings. Egan and Mullin (2012) find that warm anomalies *increase* belief in global warming. The discrepancy is explained by the agricultural interaction: the U.S. studies reflect a predominantly urban, non-agricultural population for whom weather is salient as a climate signal. India’s population is predominantly agricultural, for whom weather is salient as an economic input. The interaction model implies that for states with agricultural shares below approximately 0.49, the net effect of warming on climate search interest is positive—consistent with the U.S. findings. The sample-split estimate for low-agricultural states ( $-0.20$ , insignificant) does not turn positive because the binary split is a blunt tool that pools states with agricultural shares from 0.15 to 0.59. The continuous interaction specification, which more precisely maps the relationship, shows the sign reversal clearly (Figure 1).

This highlights a fundamental limitation of the existing literature: the weather-to-beliefs channel is not universal. It depends critically on the economic structure that mediates weather experiences. Extrapolating U.S. findings to developing countries without accounting for agricultural dependence would yield misleading predictions.

### 7.4 Limitations

Several limitations warrant explicit discussion. First, Google Trends measures search behavior, not beliefs. People may search for “climate change” out of curiosity, alarm, or skepticism. I

interpret the measure as “attention to climate change” rather than “belief in climate change,” following [Herrnstadt and Muehlegger \(2014\)](#).

Second, the agricultural share interaction captures state-level heterogeneity in a blunt way. States differ in many dimensions beyond agriculture, and the interaction could reflect urbanization, education, media access, or other correlated factors. I cannot isolate the agricultural channel from these correlated characteristics.

Third, the analysis uses one weather station per state (the state capital), which may not represent weather conditions across the entire state, particularly for geographically large states like Rajasthan or Madhya Pradesh. Spatial measurement error is likely attenuated by taking monthly averages but remains a limitation.

Fourth, the Bartik IV has a weak first stage in the single-instrument specification, limiting its informativeness. The OLS reduced-form remains the primary specification.

Fifth, the sample of 22 states with 240 monthly observations each provides reasonable variation for the primary specification, but the effective number of clusters (22 states) is small for asymptotic inference with state-clustered standard errors. [Egan and Mullin \(2012\)](#) face similar challenges with 50 U.S. states. The  $p$ -values I report should be interpreted with this caveat in mind; a more conservative reader might require  $p < 0.01$  for the interaction coefficient, which the log specification achieves.

Finally, the study period (2004–2023) coincides with India’s rapid digitization, which means the composition of the Google-searching population changes substantially over time. In 2004, internet users were overwhelmingly urban, young, and affluent; by 2023, smartphone penetration had extended digital access to semi-rural areas. The state and year fixed effects absorb average changes in digital composition, but to the extent that digitization changes the *responsiveness* of the Google-searching population to weather, there could be time-varying selection effects. The post-2010 robustness check addresses this partially, but the concern remains.

## 8. Conclusion

When the monsoon fails in Bihar, people think about their rice crop. When temperatures spike in Bangalore, people Google “climate change.” This paper presents suggestive evidence that the effect of extreme weather on climate search interest in India varies systematically with agricultural structure. In states with low crop dependence, temperature anomalies are associated with increased climate-related search interest—consistent with the experiential learning mechanism documented in the U.S. and European literature. In highly agricultural states, the same weather shocks are associated with *decreased* climate searches, a pattern

consistent with economic urgency displacing abstract environmental concern. The differential pattern is robust across specifications, though the small number of state clusters (22) limits statistical precision under wild cluster bootstrap inference.

If the pattern reflects a genuine causal mechanism, the implication for climate policy is important: building public awareness of climate change may not be able to rely on the passive channel of weather experience alone. In agricultural societies—which include the majority of the world’s most climate-vulnerable populations—weather shocks may be processed through an economic lens that does not naturally connect to the global warming narrative. Effective climate communication may need to bridge this gap explicitly.

The finding also suggests that as countries develop and agriculture’s economic share declines, weather experience may become a more potent channel for climate awareness—a structural transformation dividend that has not been previously recognized. India is on this trajectory: agriculture’s share of GDP has fallen from 23 percent in 2000 to 17 percent in 2023, and the share of the labor force in agriculture continues to decline. As this structural transformation proceeds, the cross-state variation in agricultural exposure that drives my results will diminish, and the U.S.-style experiential learning channel may strengthen. A longitudinal study tracking this transition would be a natural extension.

Several directions for future research emerge. First, survey-based evidence from Indian farmers and urban workers would help distinguish the mechanisms I propose—cognitive framing, information environment, and opportunity cost of attention—which the Google Trends data cannot separate. Second, extending the analysis to other developing countries with different agricultural structures (Sub-Saharan Africa, Southeast Asia) would test whether the agricultural suppression effect generalizes beyond India. Third, examining whether climate-related media coverage (as opposed to search behavior) shows the same agricultural heterogeneity would help disentangle supply-side and demand-side explanations for the pattern. Fourth, the interaction of extreme weather events with climate communication interventions—do targeted information campaigns in agricultural areas mitigate the suppression effect?—is a policy-relevant question that this paper motivates but cannot answer.

## Acknowledgements

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**Project Repository:** <https://github.com/SocialCatalystLab/ape-papers>

**Contributors:** @ailscl

**First Contributor:** <https://github.com/ai1scl>

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## A. Data Appendix

### A.1 Google Trends Data Collection

Google Trends data were collected using the `gtrendsR` R package (version 1.5.1). For each of 22 Indian states, I queried monthly search interest from January 2004 to December 2023 for four terms: “global warming,” “climate change” (climate terms), “cricket,” and “Bollywood” (placebo terms). The geographic parameter uses Google’s ISO 3166-2:IN codes (e.g., IN-MH for Maharashtra, IN-KA for Karnataka).

Google Trends returns a relative search interest index scaled to  $[0, 100]$  within each geographic unit and time range, where 100 represents the peak of search interest. Values reported as “<1” are set to 0.5. The climate search index averages the two climate terms; the placebo index averages the two placebo terms.

### A.2 NASA POWER Weather Data

Daily temperature and precipitation data were obtained from the NASA POWER database (version 2.3.14), which provides satellite-derived surface meteorology at  $0.5^\circ$  resolution. The `T2M` (2-meter air temperature,  $^\circ\text{C}$ ), `T2M_MAX` (daily maximum temperature), and `PRECTOTCORR` (bias-corrected total precipitation, mm/day) parameters were queried for the latitude-longitude coordinates of each state capital.

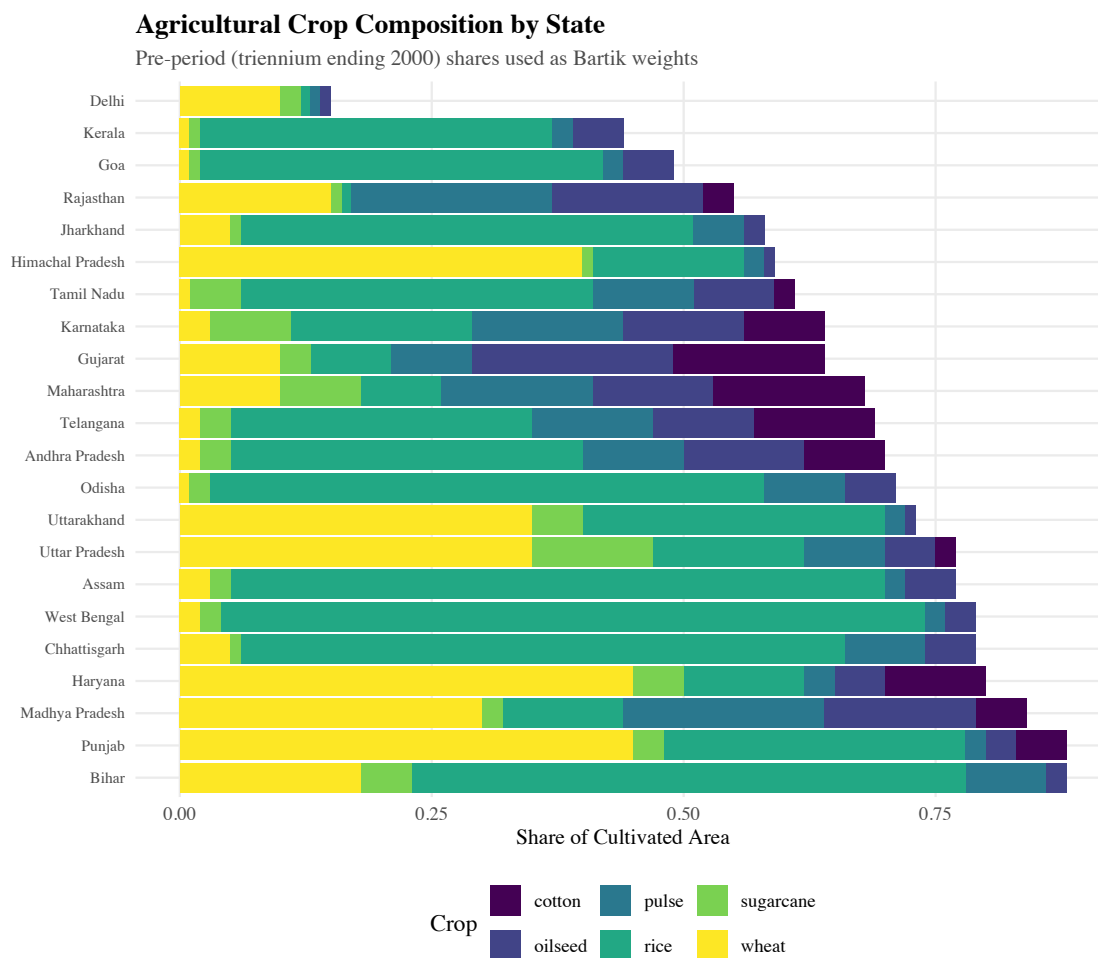
Data were downloaded for three periods: 1981–2000 (climate normals baseline), 2001–2020, and 2021–2023, using the `/temporal/daily/point` API endpoint with `community=AG`. Daily values flagged as missing (–999) were excluded.

Monthly aggregation computed the mean of daily temperatures and the sum of daily precipitation within each state-month cell. Climate normals were computed as the 1981–2000 average for each state  $\times$  calendar month cell. Anomalies are deviations from these normals.

### A.3 Agricultural Crop Shares

Pre-period crop area shares were compiled from published data of the Indian Ministry of Agriculture’s “Agricultural Statistics at a Glance” and the ICRISAT District Level Database. Values represent the approximate share of gross cropped area under each crop group for the triennium ending 2000 (i.e., the agricultural years 1997–98 through 1999–2000). Six crop groups are used: rice, wheat, pulses, oilseeds, cotton, and sugarcane.

## A.4 State-Level Crop Composition

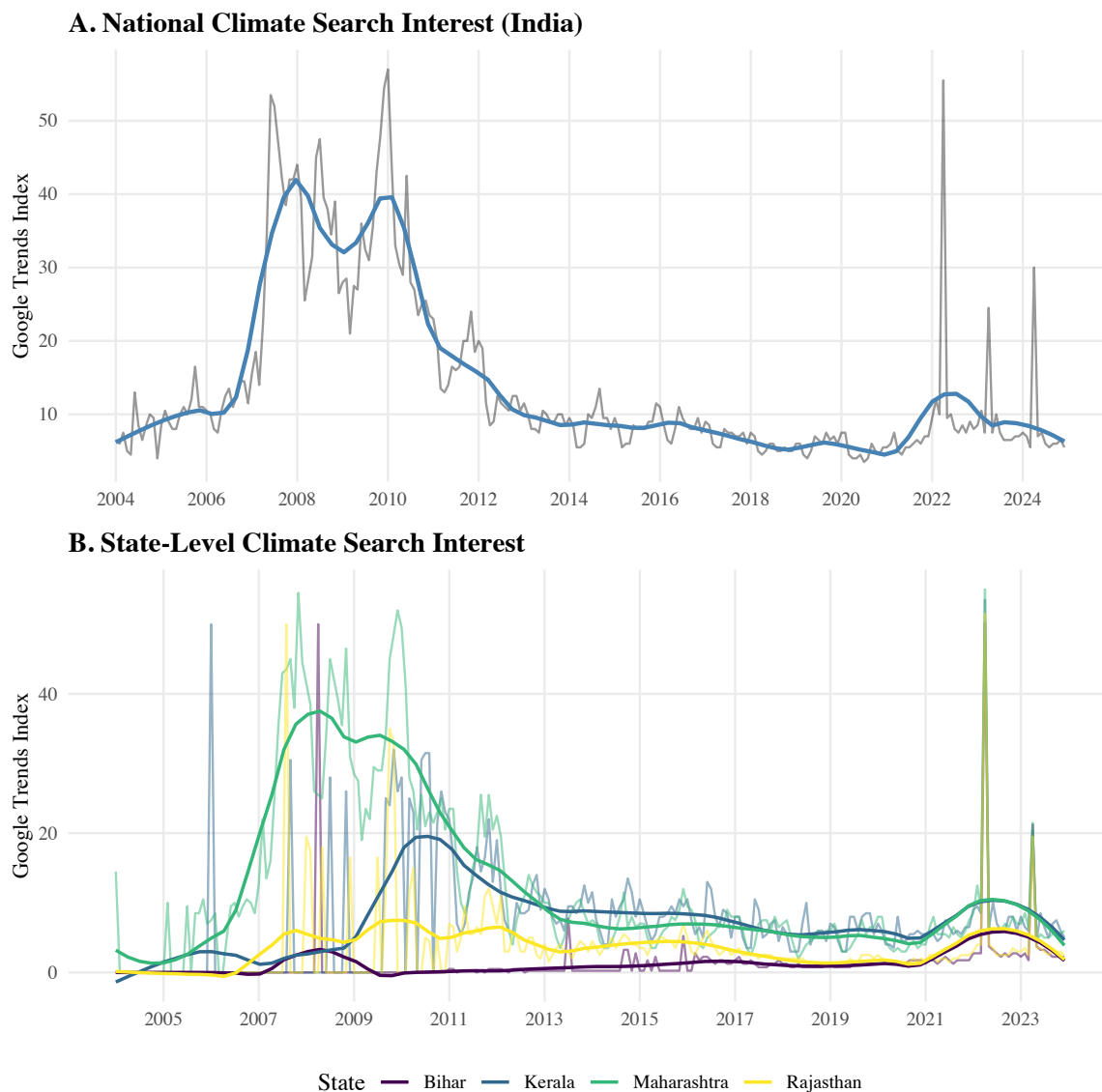


**Figure 5:** Agricultural Crop Composition by Indian State

*Notes:* Pre-period (triennium ending 2000) shares of gross cropped area under six major crop groups. Source: Ministry of Agriculture and ICRISAT District Level Database.

## B. Additional Results

### B.1 Google Trends Time Series



**Figure 6:** Climate Search Interest Over Time: National and Selected States

*Notes:* Panel A shows national-level Google Trends index for “global warming” and “climate change” averaged. The raw Google Trends query returns data through the query date; the regression sample uses January 2004 to December 2023 only. Panel B shows the same for four illustrative states: Maharashtra (high internet, moderate agriculture), Kerala (high internet, moderate agriculture), Bihar (low internet, high agriculture), and Rajasthan (low internet, high agriculture).

## B.2 Robustness Summary

Table 6 summarizes the seven robustness checks on the agricultural interaction described in the text. Five of seven specifications yield negative and statistically significant coefficients at the 5 percent level: R2 (adding precipitation), R4 (state trends), R7 (non-monsoon), R8 (post-2010), and R9 (year FE). The monsoon-only sample (R6) shows no differential effect—the interaction coefficient is small and positive—indicating that the negative agricultural-state response is driven entirely by non-monsoon months. Excluding Delhi (R5) produces a larger point estimate ( $-1.507$ ) but inflates the standard error due to reduced cluster count, rendering it insignificant.

**Table 6:** Robustness of the Agricultural Interaction

| Spec.    | Description                        | Temp. $\times$ Ag. Share | SE      | $N$   | Notes                            |
|----------|------------------------------------|--------------------------|---------|-------|----------------------------------|
| Baseline | Full sample, state + month-year FE | $-1.198^{**}$            | (0.481) | 5,280 | Table 2, col. (4)                |
| R2       | Combined weather + interactions    | $-1.396^{**}$            | (0.517) | 5,280 | Adding precipitation interaction |
| R4       | State-specific linear trends       | $-1.190^{**}$            | (0.502) | 5,280 | Nearly identical                 |
| R5       | Excluding Delhi                    | $-1.507$                 | (1.062) | 5,040 | Point est. larger; SE inflated   |
| R6       | Monsoon season only                | 0.299                    | (0.514) | 1,760 | No diff. effect in monsoon       |
| R7       | Non-monsoon season                 | $-1.767^{***}$           | (0.564) | 3,520 | Strongest in off-season          |
| R8       | Post-2010 only                     | $-1.014^{**}$            | (0.381) | 3,696 | Stable in smartphone era         |
| R9       | Year FE (less restrictive)         | $-1.164^{**}$            | (0.420) | 5,280 | Similar magnitude                |

*Notes:* All specifications include state fixed effects and cluster standard errors at the state level. Standard errors in parentheses, clustered at the state level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .